

AP CSP | Lesson 01: Bits and Binary

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Work without a calculator. Unless stated otherwise, binary values are unsigned integers.

1. A record contains three bytes of measurement data and one additional status bit. Before any padding is added, how many bits does the record contain? [1 mark]

A. 4 B. 11 C. 24 D. 25

Answer: _____

2. A maintenance device displays the binary integer 00101101. Which decimal value does it represent? [1 mark]

A. 43 B. 45 C. 46 D. 53

Answer: _____

3. A display must encode decimal 37 as an eight-bit nonnegative binary integer. Which pattern should it use? [1 mark]

A. 00100101 B. 00100110 C. 00101001 D. 00110101

Answer: _____

4. A system uses all six-bit patterns to represent consecutive nonnegative integers beginning at 0. Which option correctly gives (number of distinct patterns, largest represented integer)? [1 mark]

A. (32, 31) B. (64, 64) C. (64, 63) D. (63, 63)

Answer: _____

5. A science fair needs unique fixed-length binary identifiers for 27 exhibits. No patterns are reserved. Which TWO statements are correct? Select exactly TWO answers. [1 mark]

A. Five bits provide exactly 31 distinct identifiers.	B. Five is the minimum number of bits needed for each identifier.
C. Changing from five bits to six bits adds only two possible identifiers.	D. Changing from five bits to six bits doubles the number of possible identifiers.

Answer: _____

6. In situation I, a fixed-width integer field receives a value larger than its maximum representable value. In situation II, a program represents one third using the finite approximation 0.3333. Which classification is correct? [1 mark]

A. I: roundoff; II: overflow	B. I: overflow; II: roundoff
C. Both situations are overflow.	D. Neither situation involves a limitation of number representation.

Answer: _____

AP CSP | Lesson 01: Bits and Binary

Written responses | Questions 7-9

7. [4 marks]

A weather station sends the pattern 00010110.

(a) State its length in bits and in bytes.

(b) Interpret the pattern as a nonnegative binary integer. Show the place-value sum and give its decimal value.

(c) Could software instead use this same pattern to identify a weather category? Explain what determines its meaning.

8. [4 marks]

A robotics log records decimal 58.

(a) Show how to decompose 58 into powers of two and write its binary representation without leading zeros.

(b) Write the same value using exactly eight bits.

(c) Which is greater: your eight-bit result or the binary integer 00111100? Justify your comparison.

9. [4 marks]

A controller uses seven-bit codes.

(a) How many distinct patterns are possible? Show the expression.

(b) If interpreted as nonnegative binary integers, what is the largest value?

(c) Codes 0000000 and 1111111 are reserved. How many codes remain for actions?

(d) Can the controller assign distinct codes to 125 actions? State how many available action codes would remain unused.

AP CSP | Lesson 01: Bits and Binary

Written responses | Questions 10-12

10. [4 marks]

An aquarium assigns a unique fixed-length binary code to each species. No codes are reserved.

- (a) Find the minimum number of bits needed for 41 species. Justify both why your answer works and why one fewer bit fails.
- (b) The collection grows to 100 species. Find the new minimum code length and justify it.
- (c) With your code length from (b), how many patterns remain unused?

11. [4 marks]

A dashboard changes its numeric tags from seven bits to nine bits.

- (a) By what factor does the number of possible patterns increase? Show why.
- (b) An old tag is 0011011. Proposal X extends it to 000011011; proposal Y extends it to 001101100. Which proposal preserves the original nonnegative integer value? Verify your answer by finding the decimal values of the old tag and both proposals.

12. [4 marks]

- (a) A four-bit field represents nonnegative binary integers. A device tries to store 16. Name and explain the possible problem.
- (b) Would eight bits prevent integer overflow forever? Explain.
- (c) A program approximates $1/7$ as 0.142857. Name and explain this representation error.
- (d) Would using more bits guarantee exact representation of every real number? Explain.
